Greedy Algorithms

- **Activity Selection Problem**
- **Characteristics of Greedy Algorithm**
- **Elements of greedy strategy**
- **Making a Change Problem**
- **Fractional Knapsack Problem**
- **Minimum spanning Tree Algorithms**
- **The other problems (if time permits)**

Greedy algorithms

- A *greedy algorithm* always makes the choice that looks best at the moment
  - My everyday examples:
    - Driving in Baroda
    - Playing cards
    - Invest on stocks
    - Choose a university
  - The hope: a locally optimal choice will lead to a globally optimal solution
  - For some problems, it works
- greedy algorithms tend to be easier to code

An Activity Selection Problem (Conference Scheduling Problem)

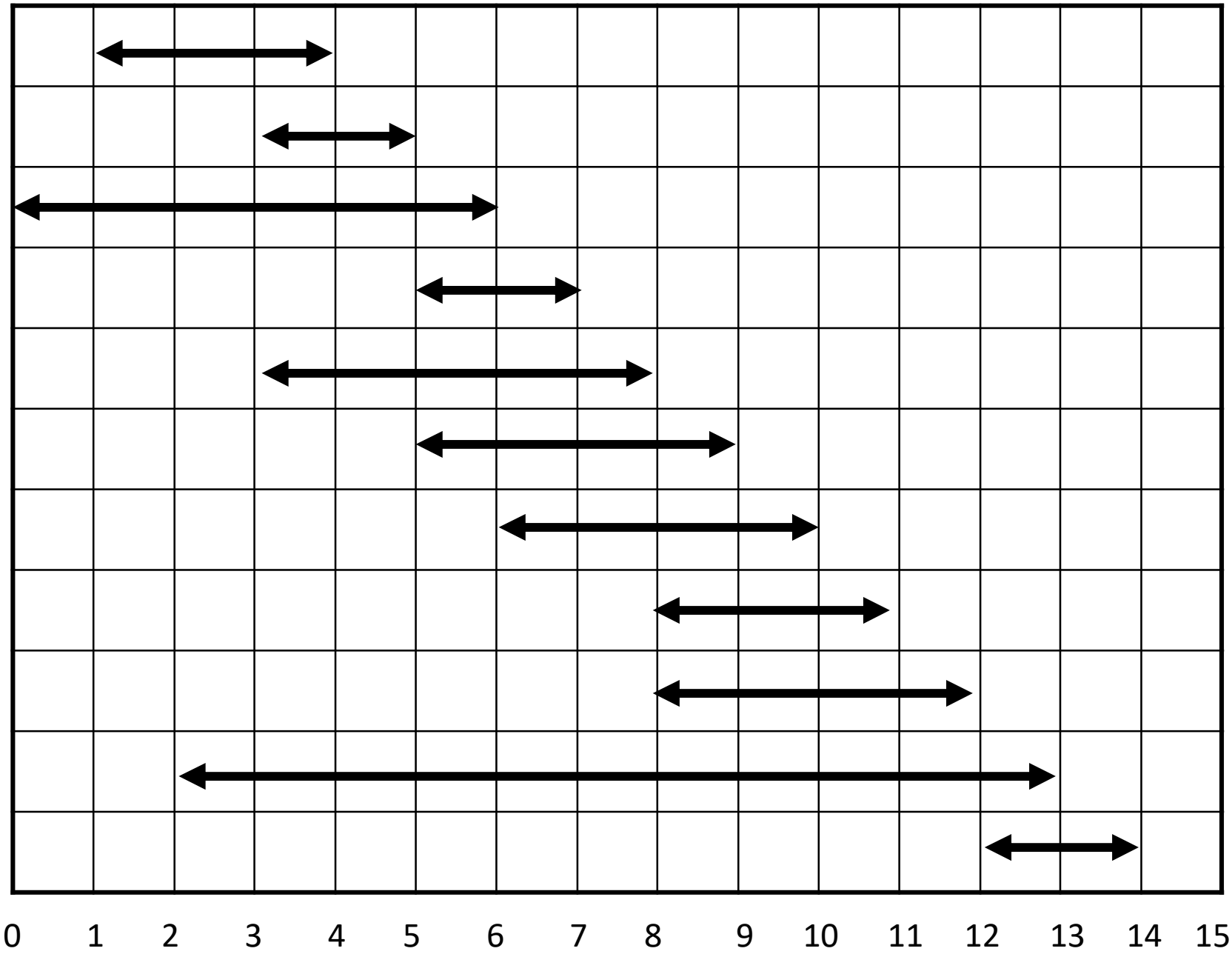
- **Input: A set of activities S = $\{a_1, \ldots, a_n\}$**
- Each activity has start time and a finish time
  - $a_i = (s_i, f_i)$
- Two activities are compatible if and only if their interval does not overlap
- **Output: a maximum-size subset of mutually compatible activities**

The Activity Selection Problem

- Here are a set of start and finish times

i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14

- What is the maximum number of activities that can be completed?
  - **$\{a_3, a_9, a_{11}\}$** can be completed
  - But so can $\{a_1, a_4, a_8, a_{11}\}$ which is a larger set
  - But it is not unique, consider $\{a_2, a_4, a_9, a_{11}\}$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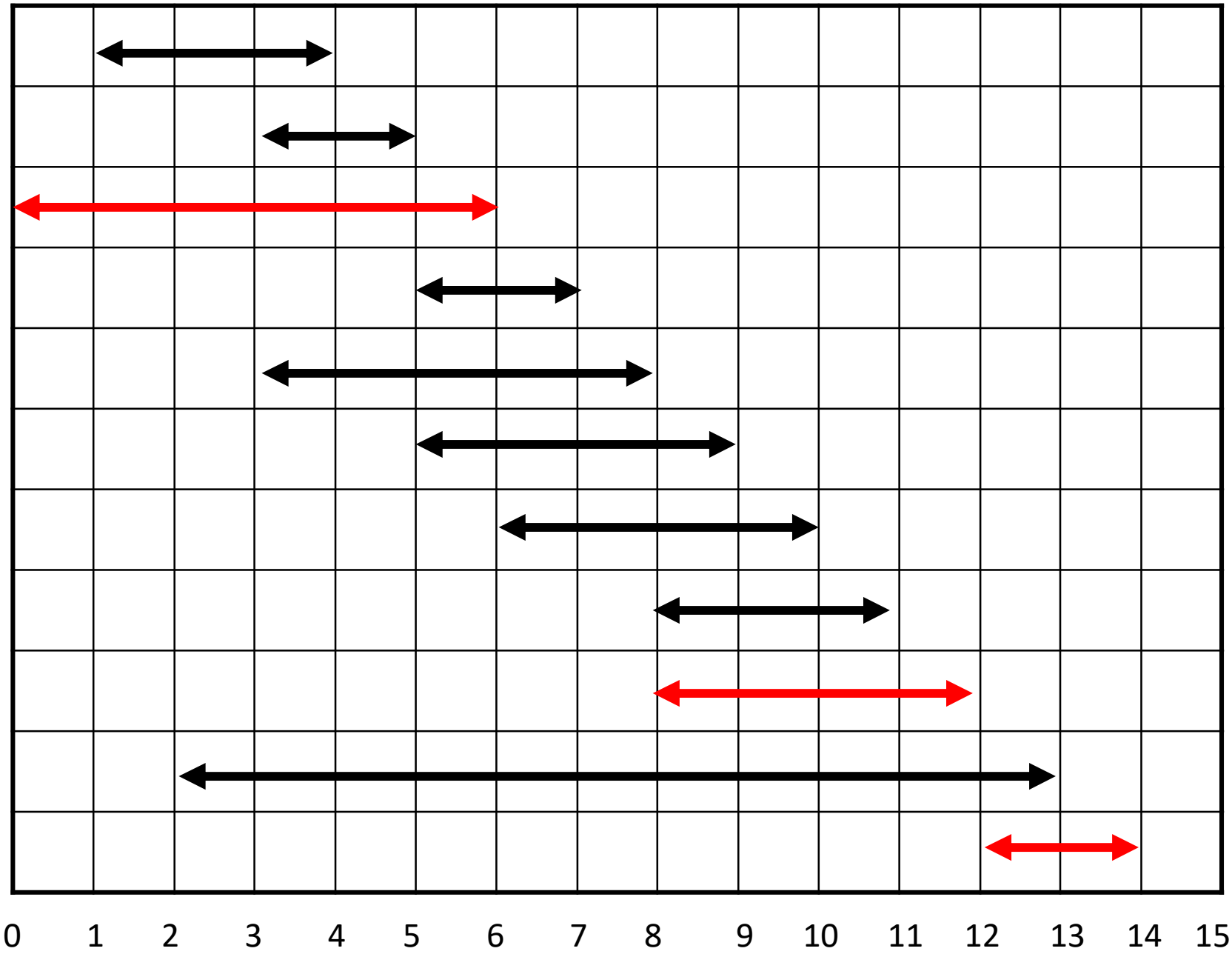
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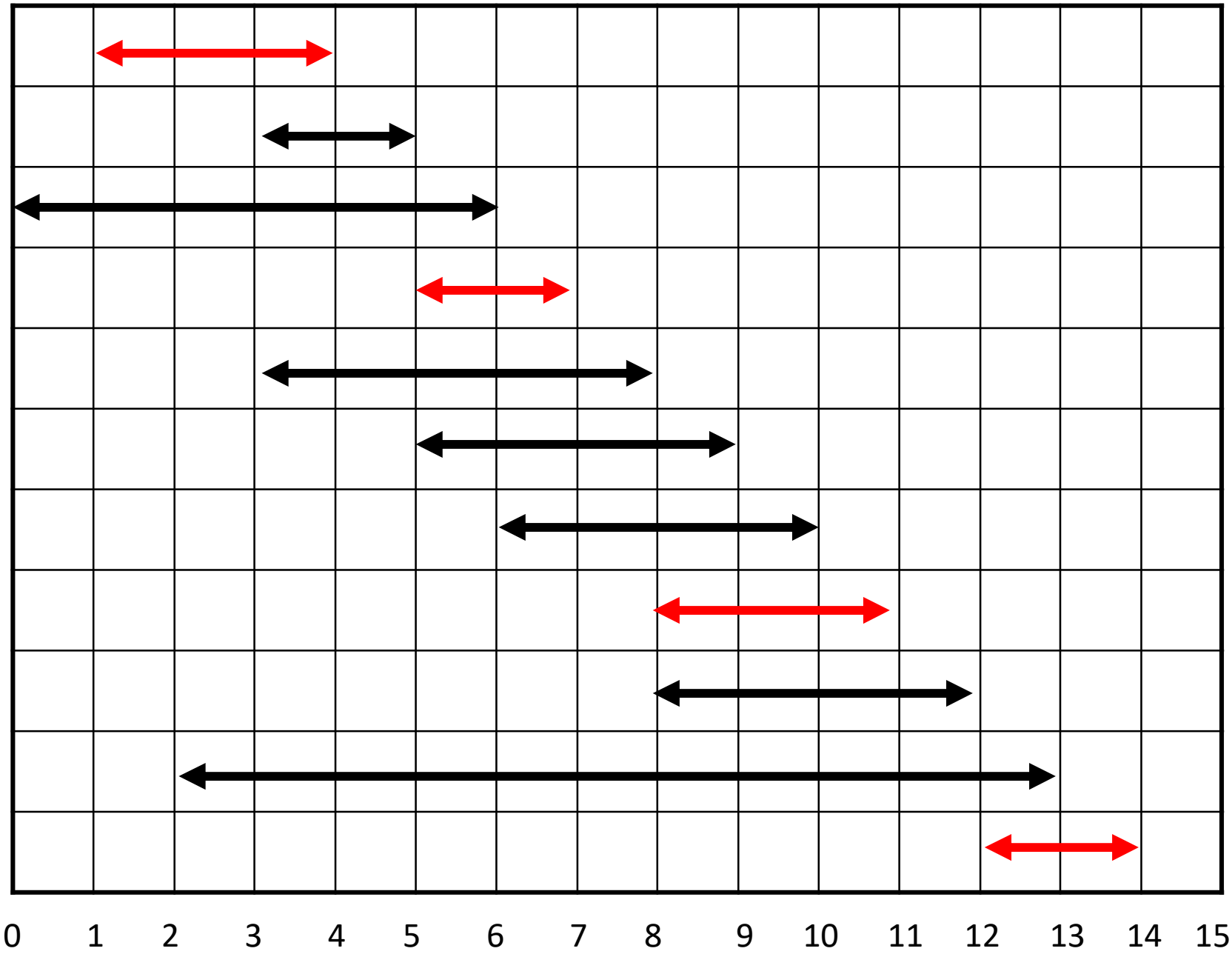
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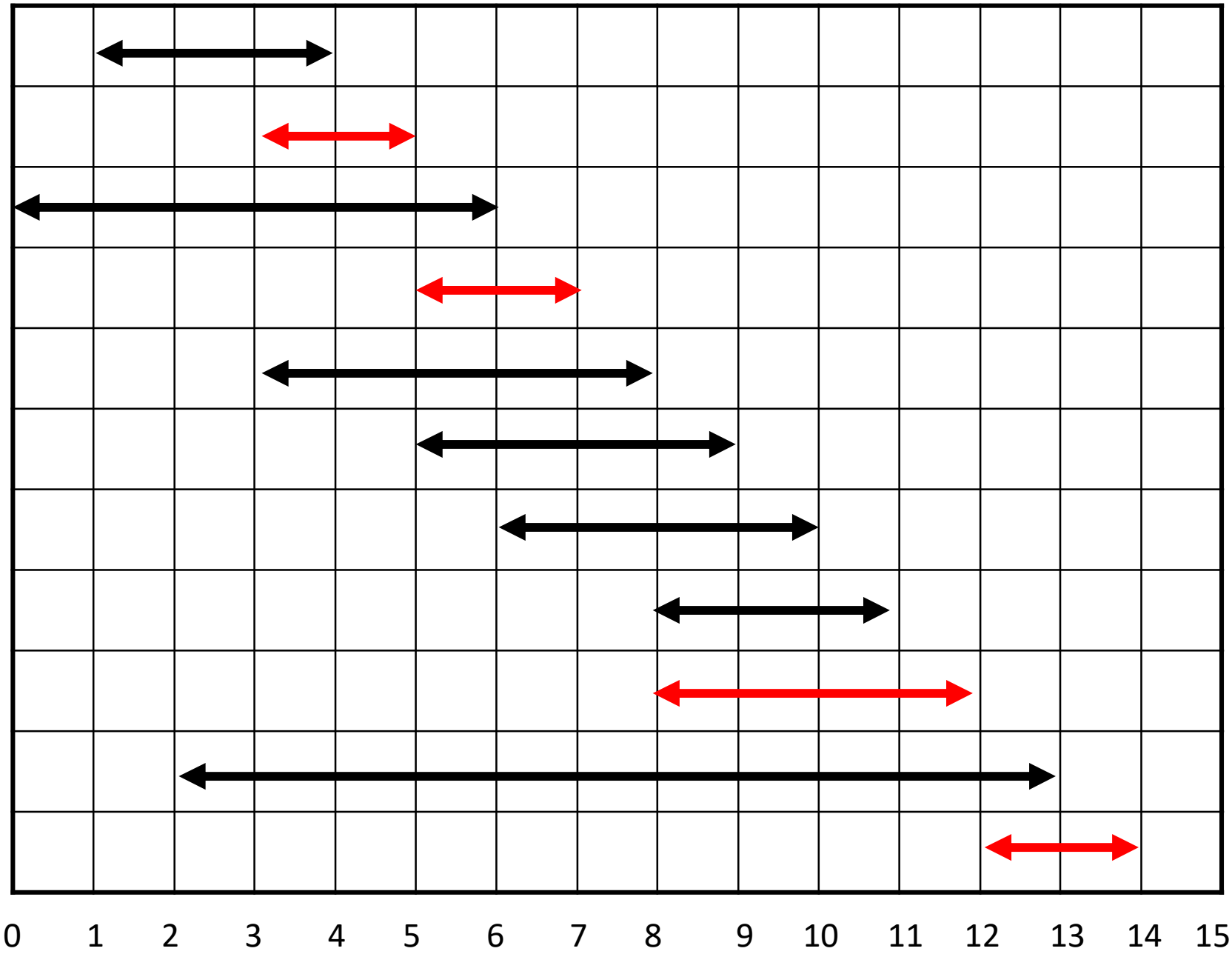
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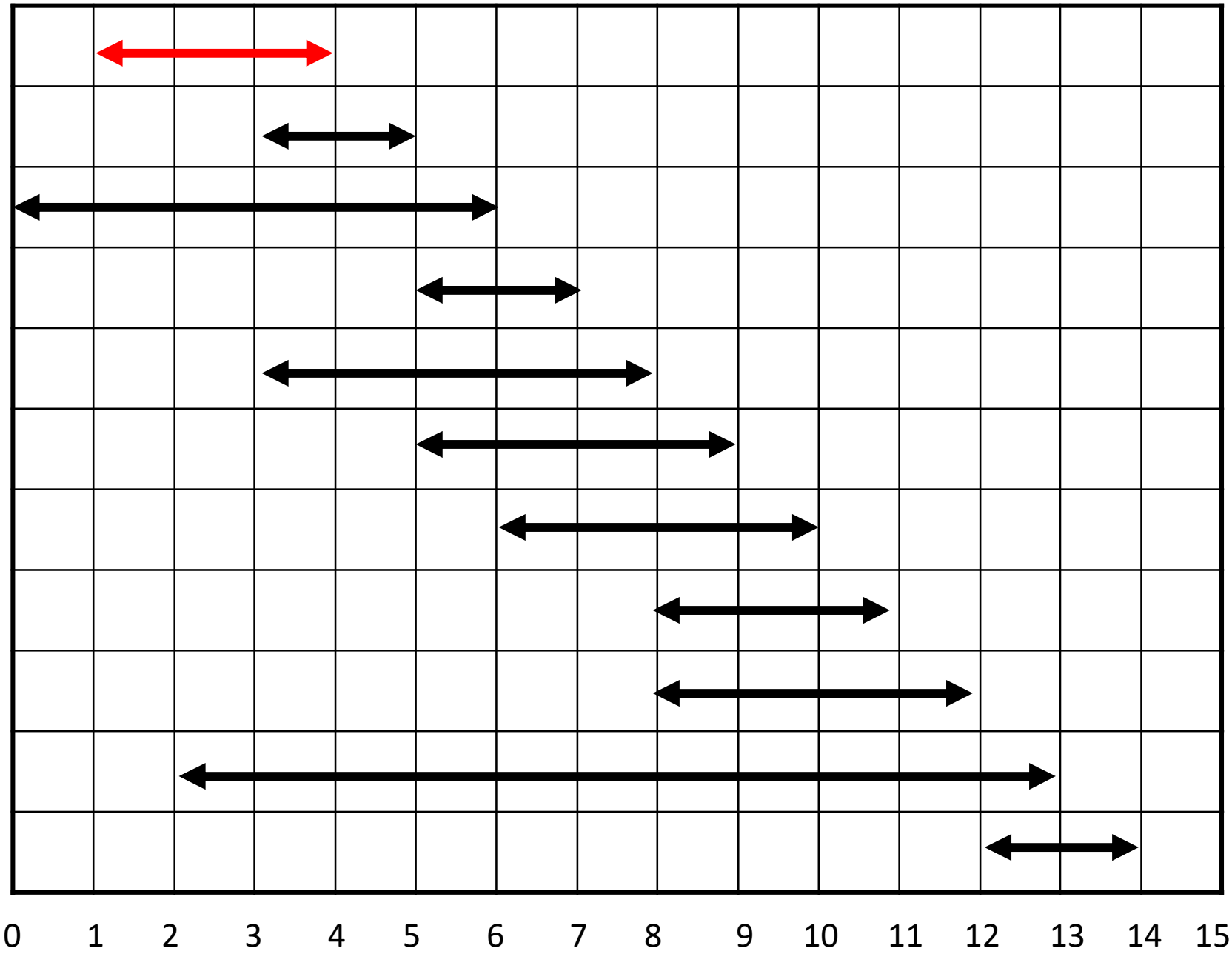
Early Finish Greedy

- Select the activity with the earliest finish
- Eliminate the activities that could not be scheduled
- Repeat!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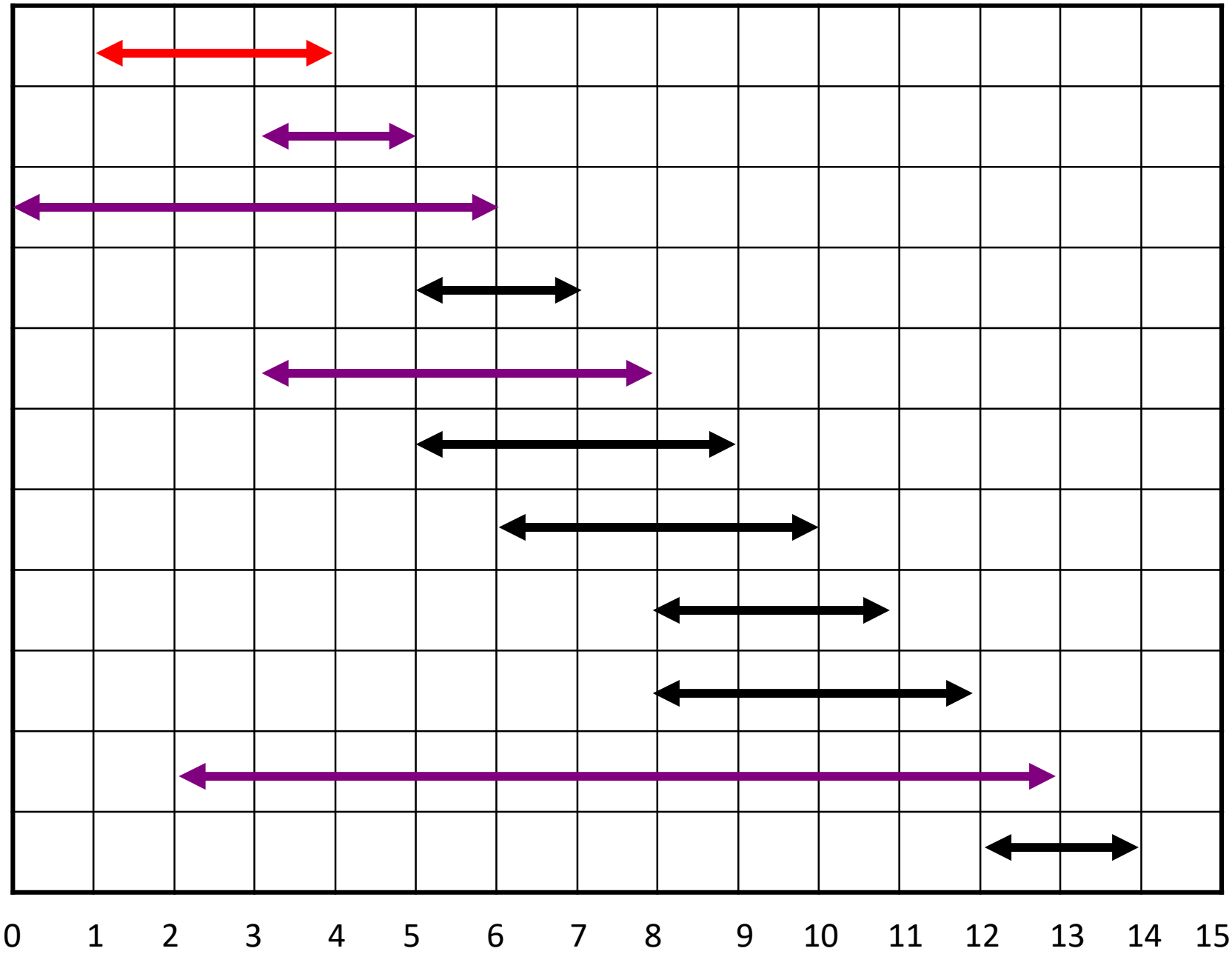
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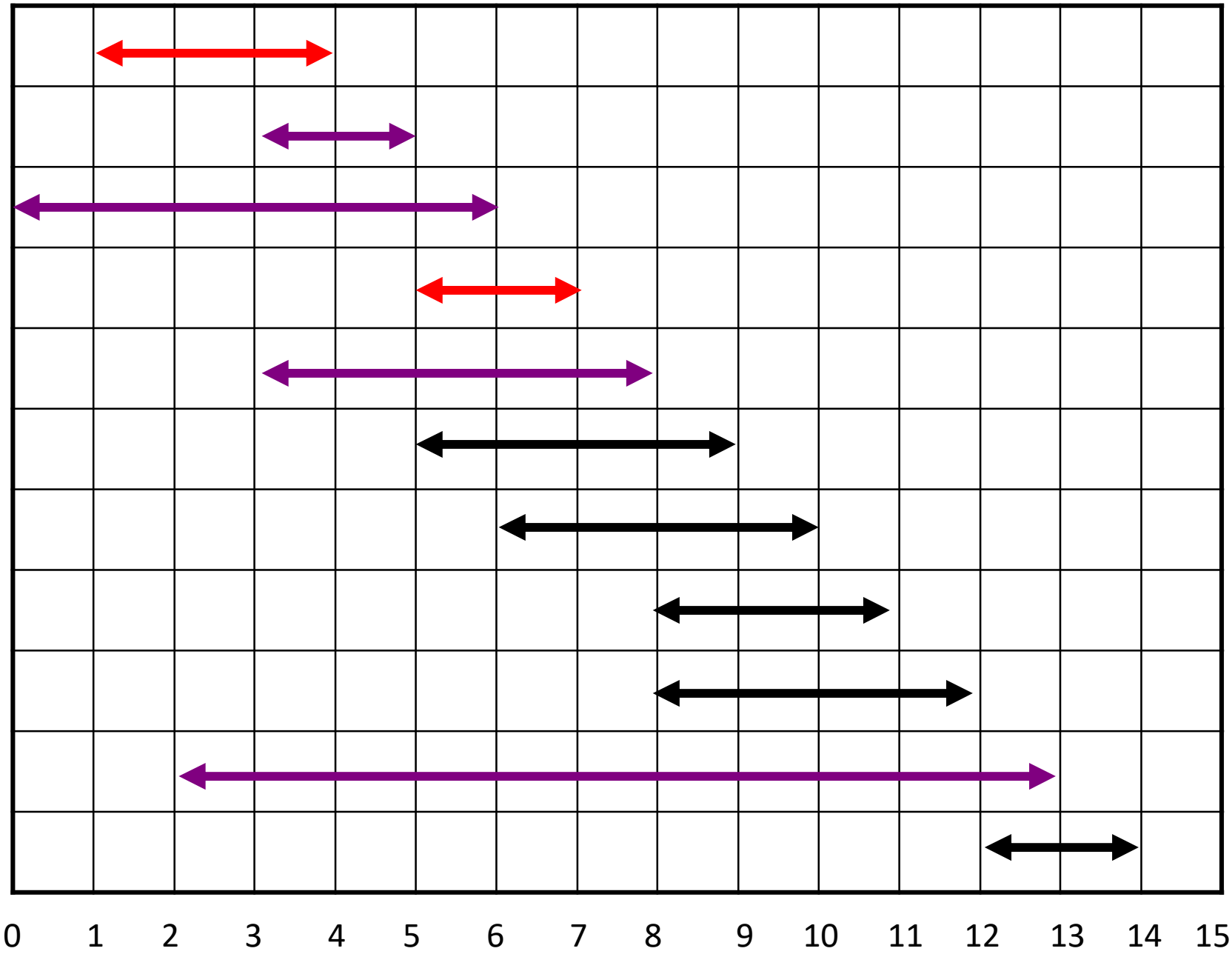
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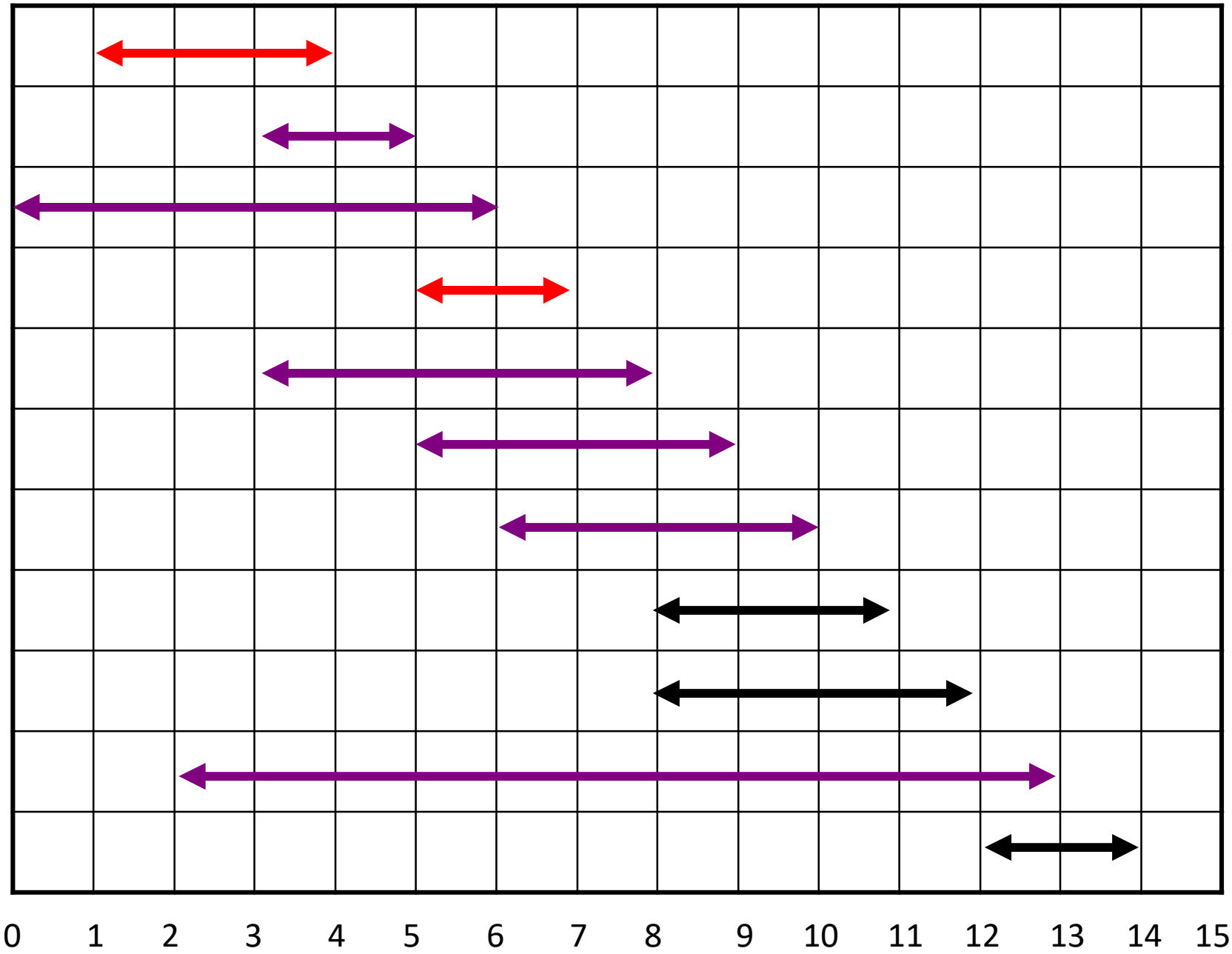
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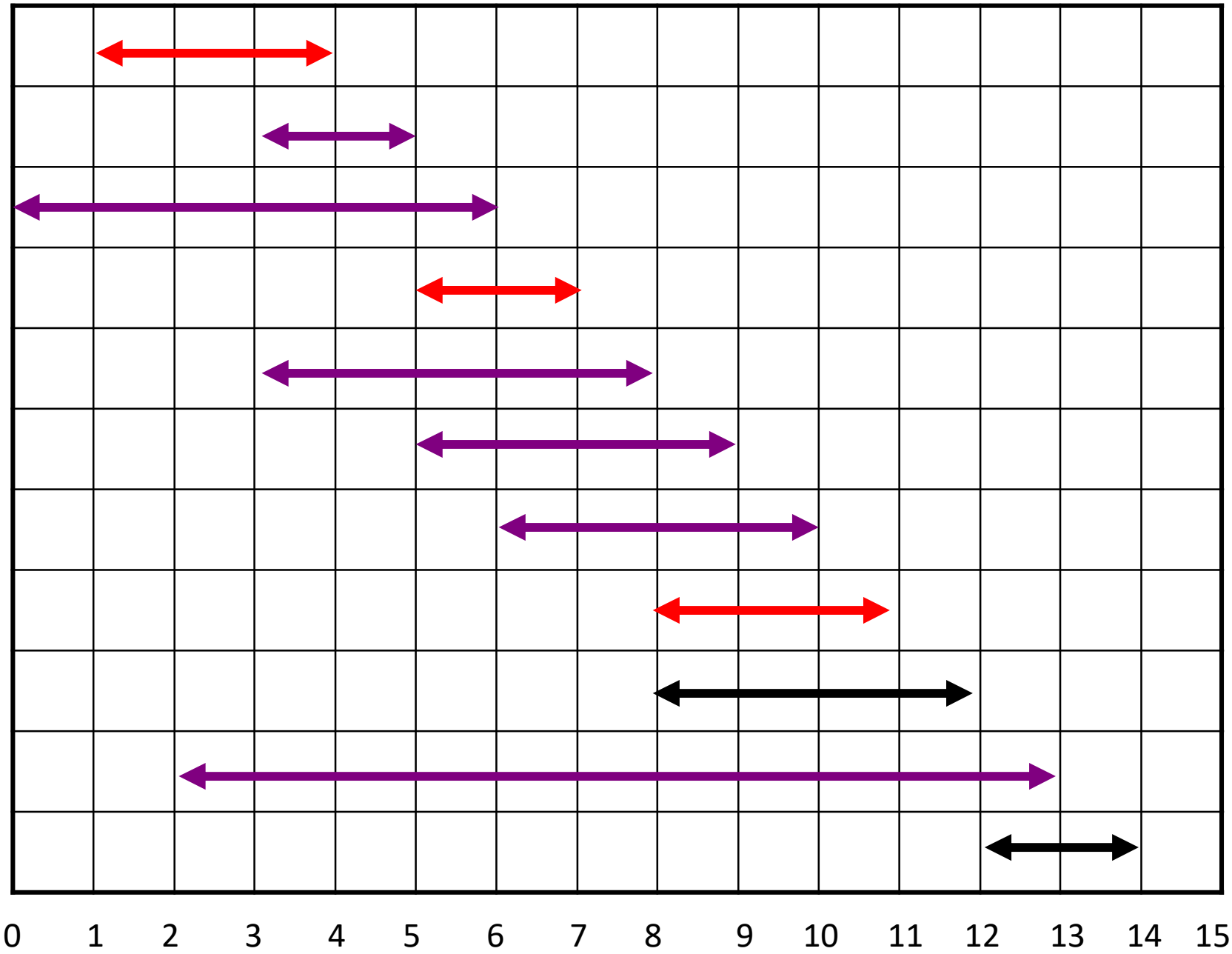
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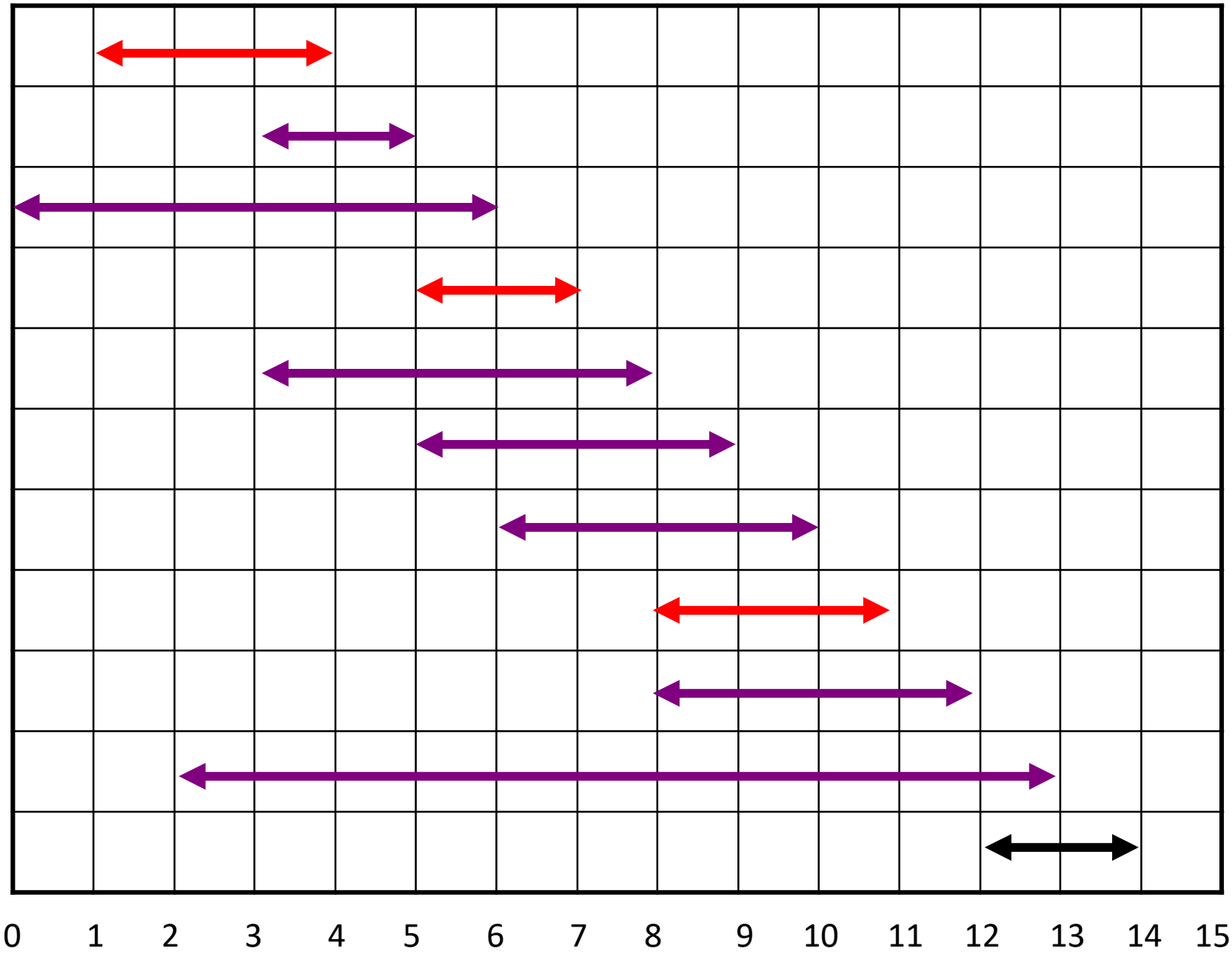
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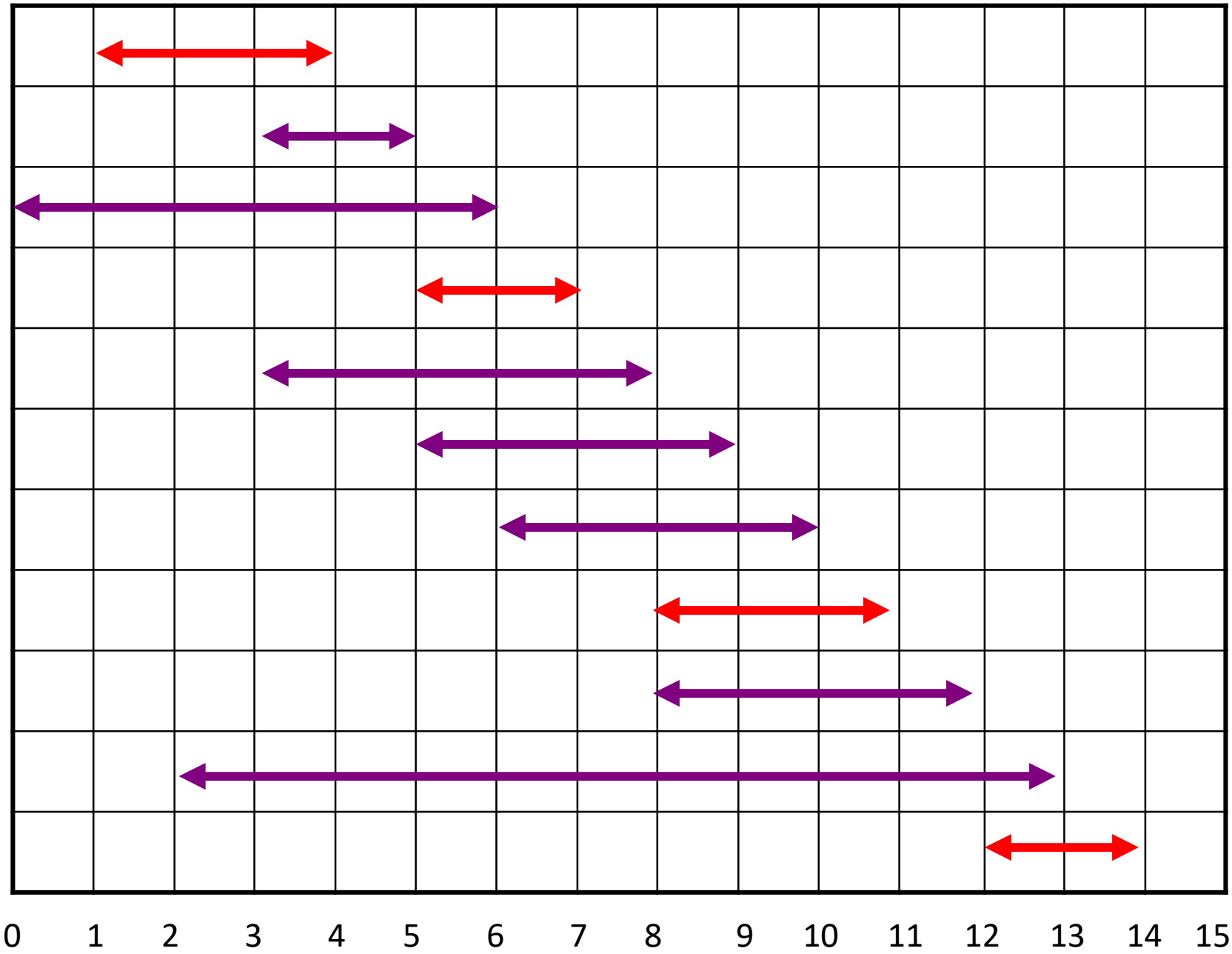
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Assuming activities are sorted by finish time

```
GREEDY-ACTIVITY-SELECTOR(s, f)
n ← length[s]
A ← {a_1}
i ← 1
for m ← 2 to n
        do if s_m ≥ f_i
                then A ← A ∪ {a_m}
                        i ← m
return A
```

Why it is Greedy?

- Greedy in the sense that it leaves as much opportunity as possible for the remaining activities to be scheduled
- The greedy choice is the one that maximizes the amount of unscheduled time remaining

Characteristic of Greedy Algorithm

- ***Solution function*** checks whether a particular set of candidates provides a solution to problem.(ignoring optimal solution)
- ***Feasible function*** checks whether or not it is possible to complete the set by adding further candidates so as to obtain at least one solution
- ***Selection function*** indicates at any time which of the remaining candidates is most promising
- ***Objective function*** gives value of the solution

Characteristic of Greedy Algorithm

- **Greedy(C: Candidate set)**

```
S← Φ
While C != Φ and not solution(S)
        x← select(C)
         C= C / {x}
         if(feasible(S U{x}) then S← S U {x}
end
If solution(S)
   return S
else
       return "No Solution found"
end
```

Elements of Greedy Strategy

- An greedy algorithm makes a sequence of choices, each of the choices that seems best at the moment is chosen
  - NOT always produce an optimal solution
- Two ingredients that are exhibited by most problems that lend themselves to a greedy strategy
  - Greedy-choice property
  - Optimal substructure

Greedy-Choice Property

- A globally optimal solution can be arrived at by making a locally optimal (greedy) choice
  - Make whatever choice seems best at the moment and then solve the sub-problem arising after the choice is made
  - The choice made by a greedy algorithm may depend on choices so far, but it cannot depend on any future choices or on the solutions to sub-problems
- Of course, we must prove that a greedy choice at each step yields a globally optimal solution

Optimal Substructures

- A problem exhibits optimal substructure if an optimal solution to the problem contains within it optimal solutions to sub-problems